Making HEPA's bound empirical: an a priori test for whether HEPA will work on a new dataset

Aviral Garg · Forgis research proposal · 10 October 2026 · on HEPA (Petersen et al., arXiv:2605.11130v4)

The gap I want to close, stated by the paper itself

Proposition 1: $I(H_t;E) \geq I(H^*;E) - C_\eta L^2 \varepsilon$, with vacuity threshold (eq. 17) $\varepsilon_{\text{vac}} = I(H^*;E)/(C_\eta L^2)$.

- Within a dataset it is validated: varying ε via pretraining snapshots gives Spearman $\rho = -0.87$ / -0.67 / -0.64 / -0.49 on C-MAPSS-1 / C-MAPSS-3 / MBA / SMAP.
- Across datasets it falls apart. Pooling converged ε over the 14 benchmarks gives Pearson $r = -0.05$ ($p = 0.90$). The paper attributes this to L, C_η and target scale differing by dataset.
- The constants are never estimated. In the authors' words: "The constants L and C_η are not estimated directly; fig. 3 validates only the monotonic relationship, not the full quantitative bound," and in Future Work: "deriving fully empirical versions of the information-retention bound that estimate L and C_η directly from data remains an interesting open problem."

As it stands the bound only explains HEPA's failures after the fact. I want to make it predictive. Define the dimensionless slack

$$\hat{\rho}(\mathcal{D}) \;=\; \frac{\hat{\varepsilon}}{\hat{\varepsilon}_{\text{vac}}} \;=\; \frac{\hat{\varepsilon}\cdot \hat{C}_\eta \hat{L}^2}{\hat{I}(H^*;E)}, \qquad \text{bound non-vacuous} \iff \hat{\rho} < 1.$$

Hypotheses

- H1 (normalisation restores the cross-dataset signal). Across the 14 benchmarks, $\rho_s(\hat{\rho},\ \text{h-AUROC}) \leq -0.6$, where raw ε gives $r = -0.05$. Falsified if $|\rho_s| < 0.3$, which would mean the incommensurability comes from somewhere else and my normalisation is wrong. The objection I have to design against: $\hat{I}(H^*;E)$ comes from a probe fitted on the same labels h-AUROC uses, so part of any correlation could be the probe predicting itself. Control: report the partial correlation of $\hat{\rho}$ with h-AUROC given the probe's own AUROC, on disjoint rows. If it vanishes there, H1 is dead and I would say so.
- H2 (it works as a screen). A prediction: I have not computed $\hat{\rho}$ anywhere yet. $\hat{\rho} \geq 1$ should mark where the bound gives no guarantee. Pre-registered: CHB-MIT, excluded at h-AUROC 0.497 with $I(H^*;E) \approx 0$ (A.6), lands at $\hat{\rho} \geq 1$; C-MAPSS-1 (0.806) at $\hat{\rho} \ll 1$. CHB-MIT alone is a weak test: the paper already reports $I(H^*;E) \approx 0$ there, so $\hat{\rho} \geq 1$ is near-arithmetic. The real test is the ordering over all 14, i.e. H1. H2 falsified if either control is misclassified, or if the threshold needs per-dataset tuning. What it does not mean: appendix A.4 says a vacuous bound does not imply $I(H_t;E) = 0$, and GECCO proves it: within-dataset $\rho = +0.14$ ($p = 0.67$), yet HEPA scores 0.88 there, among its strongest table-1 results. The screen is about the guarantee, never a prediction that HEPA will fail.
- H3 (the L1/L2 gap costs more than monotonicity). A2 is in L2 while eq. (2) trains L1. Appendix A.2 already answers this via $\|u\|_2 \leq \|u\|_1$, calling L1 a "monotone proxy for ε", which is enough for the within-dataset test in fig. 3 but not here, since $\hat{\rho} \geq 1$ needs ε on its scale rather than its ordering. Prediction: $\hat{\rho}$ from the measured L2 residual beats the L1 version by $\Delta\rho_s \geq 0.15$. Falsified if they sit within noise, which would license the L1 loss.

Estimators, all cheap and low-dimensional where it matters

Fit HEPA's own event head $\hat{\eta}(h) = \sigma(w^\top \mathrm{LN}(h) + b)$ (LayerNorm + linear, 769 params on d=256, their appendix C) on the frozen target representations h^*. Then, on held-out data:

- $\hat{L} = \|\gamma \odot w\|_2/(4\hat{s}_{\min})$, with γ the LayerNorm gain and $\hat{s}_{\min}$ the smallest held-out $\sqrt{\mathrm{Var}(h^*) + \epsilon}$. The LayerNorm Jacobian is easy to drop and costly to drop: $\nabla_h \hat{\eta} = \sigma'(z) s^{-1} P(\gamma \odot w)$, so $\|w\|_2/4$ is neither exact nor an upper bound. On the unit sphere at d=256, $s \approx 1/16$, so omitting it understates L by $\sim$16× and $\hat{\rho}$ by $\sim$250×, toward calling a vacuous bound non-vacuous.
- $\hat{C}_\eta = \big(2\,\underline{\eta}\,(1-\overline{\eta})\big)^{-1}$ with $\underline{\eta}, \overline{\eta}$ the 1st/99th percentiles of $\hat{\eta}(h^*)$.
- $\hat{I}(H^*;E) = H(\pi_e) - \mathbb{E}\big[H(\hat{\eta}(H^*))\big]$, plug-in. E is binary, so the entropy is 1-D and no high-dimensional MI estimator is needed. The difficulty moves rather than vanishes: this is a lower bound on I only as far as $\hat{\eta}$ fits the true posterior, so I would report it against a higher-capacity $\hat{\eta}$ as a sensitivity check.
- $\hat{\varepsilon} = \mathbb{E}\|\hat{h} - h^*\|_2^2$ on a held-out split, and in L1 as well, per H3.

Why it is worth doing, what it costs, and where it can fail

- Makes HEPA's central theorem computable and falsifiable, answering the paper's own open problem; and $\hat{\rho}$ needs only unlabelled pretraining plus a small probe, so the guarantee can be checked on a new plant before committing. Cost: a re-run of HEPA's own recipe (under a minute per dataset on one A10G) plus $O(N)$ estimators, no new model.
- Risks. Plug-in MI biases $\hat{I}$ down, pushing $\hat{\rho}$ up toward the threshold; rank correlation protects H1 but H2 cannot hide behind that, so I would calibrate the threshold on held-out folds. And n=14 is low power: pre-register direction and test, and show the per-dataset scatter.

Secondary direction (from my analysis of `embeddings.npz`)

In the attached report, with a frozen target no SIGReg weight $\alpha \in [0.01, 0.9]$ bought both prediction quality and representation rank (at α=0.1: cosine 0.8716±0.0004→0.0609±0.0044, 5 seeds). HEPA's target is never frozen: weight-shared, and robust to EMA per appendix I.3, which picks joint training "for its simplicity", so it can move toward what SIGReg wants. That regime sits outside both settings I.3 tests. SIGReg is doing two jobs: preventing collapse, and setting the target's distribution.

- H4. Split the jobs: SIGReg on the target branch, an explicit rank floor on the predictor output. Prediction: h-AUROC goes flat in α over $[0.03, 0.3]$ (range ≤ 0.02), mediated by predictor-output effective rank. Falsified if h-AUROC still varies by more than 0.02, or if the rank floor binds without moving it. Not a claim about the BATADAL/MBA gap, which the paper scopes to tokenisation.

Worth confirming: the paper reports α=0.1 (table 7) while `Forgis-Labs/HEPA` PR #3 says the table-1 runs used 0.04. I've worked from 0.1.